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East Asian contributions to Dutch traditional and western commercial chickens inferred from mtDNA analysis

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Summary

Understanding the complex origin of domesticated populations is of vital importance for understanding, preserving and exploiting breed genetic diversity. Here, we aim to assess Asian contributions to European traditional breeds and western commercial chickens for mitochondrial genetic diversity. To this end, a 365-bp fragment of the chicken mtDNA D-loop region of 16 Dutch fancy breeds (113 individuals) was surveyed, comprising almost the entire breed diversity of The Netherlands. We also sequenced the same fragment for 160 commercial birds representing all important commercial types from multiple commercial companies that together represent more than 50% of the worldwide commercial value. We identified 20 different haplotypes. The haplotypes clustered into five clades. The commonest clade (E-clade) supposedly originates from the Indian subcontinent. In addition, both in commercial chicken and Dutch fancy breeds, many haplotypes were found with a clear East Asian origin. However, the erratic occurrence of many different East Asian mitochondrial clades indicates that there were many independent instances where breeders used imported exotic chickens for enhancing local breeds. Nucleotide diversity and haplotype diversity analyses showed the influence of the introgression of East Asian chicken on genetic diversity. All populations that had haplotypes of multiple origin displayed high inferred diversity, as opposed to most populations that had only a single mitochondrial haplotype signature. Most fancy breeds were found to have a much lower within-population diversity compared to broilers and layers, although this is not the case for mitochondrial estimates in fancy breeds that have multiple origin haplotypes.

Keywords broilers, diversity, Dutch fancy breeds, layers, mtDNA D-loop.

Introduction

The hypothesis that the chicken has been domesticated from the red jungle fowl is widely accepted (West & Zhou 1988; Crawford 1990; Moiseyeva *et al.* 2003), although evidence exists (Nishibori *et al.* 2005; Eriksson *et al.* 2008) that other wild species of jungle fowl have contributed genetic material to modern chickens. In addition, different opinions have prevailed regarding the geographic region of domestication. Crawford (1990) proposed that domestication of chickens took place in the Indus valley around 2500–2100 BC. However, archaeological and genetic evidence suggests that Southeast Asia is the origin of chicken domestication (West & Zhou 1988; Akishinonomiya *et al.* 1994, 1996). In contrast, Liu *et al.* (2006), found evidence implicating multiple maternal origins of chickens centred around South and Southeast Asia, reconciling these earlier findings by providing genetic evidence for domestication of chickens to have taken place in multiple geographic locations.

Many routes have been proposed to explain the migration of domestic chickens to Europe. A review by Shahbazi *et al.* (2007) indicates that chickens were introduced to Iran from the Indus valley around 2500–2000 BC and then spread to Europe, to be introduced to Greece and Italy across the

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Aegean Sea or directly to central Europe through Scythia and Southern Russia. Other possible routes include dispersion through Iran to the Mediterranean and through China and Russia to Europe (Crawford 1995). West & Zhou (1988) suggested that North European chickens were introduced from China through Russia and not from the Indus Valley. Based on the fact that chickens in the Mediterranean were morphologically different from those found in Northern Europe, West & Zhou (1988) shared the opinion that Mediterranean chickens could have been introduced through Iran.

Dispersal of domestic chickens from their putative centres of domestication to different regions with diverse environmental conditions and people of different cultures has contributed to the observed genetic differentiation of chicken populations. Other factors that may have played a role in the genetic differentiation include founder effects, migration, mutation, and natural and artificial selection. Most indigenous pure breeds and varieties and the multitude of fancy breeds found today in Europe and North America were developed during the late 19th century by breeding for exhibition traits using local and imported stocks (Crawford 1995). It was also during the last centuries that chickens from South and Southeast Asia were imported to Europe and were documented to have played a role in the formation of these European breeds.

For Dutch fancy breeds, early paintings and existing literature have shown that the formation of many breeds was influenced by Asiatic (and to a lesser extent by east European and Mediterranean) chickens. For instance, one or more of the Asiatic chicken classes such as Malays, Japanese bantams and Sumatras were involved in the formation of Friesian fowls, Dutch bantams, Breda fowls, Booted bantams, Barnevelders, Kraienkoppes and Hamburghs. Polish bearded and Polish non-bearded are thought to be influenced by chickens from east Europe. The Lakenvelders, Assendelft fowls, Drente fowls, Holland fowls and Groninger Mews are country fowls with no recorded history of genetic influence from Asiatic chickens (see Appendix S1).

From the 1950s, some of the most productive breeds and varieties that were developed by fanciers were subjected to intensive selection for quantitative traits, giving rise to the breeds currently being used in the broiler and layer industry (Muir *et al.* 2008). Broilers descend from Plymouth Rock females that have been selected for reproductive traits, while the sire lines, selected for growth traits, are based on the White Cornish breed. The Plymouth Rock has a documented origin that includes Asian breeds such as Java Fowl, Brahma and Cochin, while the Cornish are thought to be partially descended from Asian fighting cocks. Commercial egg layers consist mainly of White Leghorns or their crosses, which are bred for the production of white-shelled eggs. The brown-egg layers on the other hand are derived from the Rhode Island Red breed (Crawford 1995), which originated in the USA and is documented to include several Chinese chicken breeds in its heritage.

Therefore, both European fancy and Western commercial breeds of chicken are thought to carry genes from Asian populations of domestic chickens. However, the sources of variations and genetic relationships among fancy breeds of The Netherlands and commercial chicken populations with their potential progenitors and the probable influence of other continental populations are not clear. Up to now, studies investigating the origin of European traditional and Western commercial breeds, such as by way of mitochondrial haplotype analysis, have been limited in scope.

This study aims to elucidate possible Southeast Asian contributions to the genetic variation of European traditional breeds and western commercial chickens by ascertaining mitochondrial genetic diversity. We do this by systematically investigating the majority of chicken breeds of one European country, as well as breeds that are known to be highly diverse in origin and encompass breeds of known Asian origin, and country fowls with no known non-European contributions. In addition, we systematically investigated Western commercial populations by including all major types (white-egg layers, brown-egg layers, dam broiler and sire broiler lines), representing more than 50% of commercially sold chickens in the world. Furthermore, we compared the level of genetic diversity within and between the Dutch fancy and commercial breeds.

Materials and methods

Samples

In this study, 273 individuals were selected from populations and breeds. This included commercial chickens (n = 160) selected for economically important traits and fancy breeds (n = 113) from The Netherlands. The name and sample size for each breed are shown in Table 1.

Genomic DNA was extracted from whole blood of each chicken using the Gentra Kit (http://www1.qiagen.com/Products/GenomicDnaStabilizationPurification/GentraPureGeneBloodKit.aspx).

MtDNA D-loop sequencing

In total, 635 bp of the chicken mtDNA D-loop was amplified for 273 individuals with the primers L16750 (5′-AGGACTGCTTGAAAAGC-3′) and CR1b (5′-CCATACACGCAAACCGTCTC-3′). PCR was performed under standard conditions (annealing temperature of 50 °C). Sequencing reactions were carried out using the Big Dye terminator cycle sequencing kit version 3.1 and analysed on an ABI 3100 automated DNA analyzer. In this study, we used 365 base pairs covering the D-loop from base 30–394 of the reference mtDNA sequence of *Gallus gallus domesticus* (X52392, Desjardins & Morais 1990).

Table 1 Sample information and haplotype frequency of Dutch fancy and commercial chicken populations analysed in this study.

Breed	Abbreviation	Sample source	*n*	No. of haplotypes	Haplotype (frequency)[1]
Fancy					
Groninger Mew	GrM	Netherlands	7	3	E1 (1), E3 (1), E7 (5),
Lakenvelder	Lak	Netherlands	6	2	C1 (2), E1 (4)
Drente fowl	DrF	Netherlands	7	1	E1 (7)
Assendelft fowl	AsF	Netherlands	8	1	E1 (8),
Frisian fowl	FrF	Netherlands	5	2	A3 (2), E1 (3)
Hamburgh	Ham	Netherlands	6	2	E2 (2), E7 (4)
Polish bearded	PoB	Netherlands	6	1	E1 (6)
Dutch owl-bearded	DoB	Netherlands	6	1	E1 (6)
Polish non-bearded	PnB	Netherlands	7	1	E1 (7)
Breda fowl	BrF	Netherlands	8	3	A2 (2), D1(3), E4 (3)
Brabanter	Bra	Netherlands	8	3	E1 (5), E3 (1), E5 (2)
Dutch Bantam	DuB	Netherlands	7	3	A1 (1), E1 (5), E11 (1)
Booted bantam	BoB	Netherlands	8	2	C1 (4),E1 (4)
Barnevelder	Bar	Netherlands	8	1	E1 (8)
Holland fowl	HoF	Netherlands	8	1	E1 (8)
Kraienkoppe	Kra	Netherlands	8	1	E1 (8)
All fancy			113	12	A1 (1), A2 (2), A3 (2), C1 (6), D1 (3), E1 (80), E2 (2), E3 (2), E4 (3), E5 (2), E7 (9), E11 (1)
Commercial					
Broiler Male line 1	BM1	Company A	15	4	B1 (2), E1 (1), E5 (11), E9 (1)
Broiler Female line 2	BF2	Company A	15	4	B1 (1), E1 (1), E5 (12), E6 (1)
Broiler Male line 3	BM3	Company B	21	4	A4 (4), B1 (7), E1 (2), E5 (8)
Broiler Female line 4	BF4	Company B	42	4	A4 (2), B1 (5), E1 (2), E5 (33)
Brown-Egg Layer	BE		51	7	A1 (5), A2 (7), A5 (1), E1 (9), E3 (5), E8 (4), E10 (20)
White-Egg Layer	WE		16	4	A2 (1), E7 (7), E8 (1), E12 (7)
Broilers line 1 and 2 combined	B1–2		30	5	B1 (3), E1 (2), E5 (23), E6 (1), E9 (1)
Broilers line 3 and 4 combined	B3–4		63	4	A4 (6), B1 (12), E1 (4), E5 (41)
Combined broiler Male lines (1 & 3)	BM1–3		36	5	A4 (4), B1 (9), E1 (3), E5 (19), E9 (1)
Combined Broiler Female lines (2 & 4)	BF2–4		57	5	A4 (2), B1 (6), E1 (3), E5 (45), E6 (1)
All Broiler	B		93	6	A4 (6), B1 (15), E1 (6), E5 (64), E6 (1), E9 (1)
All Layer	L		67	9	A1 (5), A2 (8), A5 (1), E1 (9), E3 (5), E7 (7), E8 (5), E10 (20), E12 (7)
All commercial	Com		160	14	A1 (5), A2 (8), A4 (6), A5 (1), B1 (15), E1 (15), E3 (5), E5 (64), E6 (1), E7 (7), E8 (5), E9 (1), E10 (20), E12 (7)
Fancy plus commercial	Fancy-Com		273	20	A1 (6), A2 (10), A3 (2), A4 (6), A5 (1), B1 (15), C1 (6), D1 (3), E1 (95), E2 (2), E3 (7), E4 (3), E5 (66), E6 (1), E7 (16), E8 (5), E9 (1), E10 (20), E11 (1), E12 (7)

[1]Figures in parenthesis indicate the frequency of that particular haplotype

Sequence analysis

Phylogenetic relationships and network analysis

All of the 273 mtDNA D-loop sequences were manually edited and aligned with the aid of the CLUSTALX program (Thompson *et al.* 1997). A rooted neighbour-joining (NJ) tree was reconstructed to identify phylogenetic clades employing the Kimura 2-parameter model. The CLUSTALX package was used to draw the tree applying 1000 bootstrap replications to test reliability of the branching order. The red jungle fowls (*Gallus gallus bankiva*, AB007718 and *G. g. gallus*, AB007720) were used as outgroups to root the tree constructed using NJPLOT95. Median-joining (MJ) networks (Bandelt *et al.* 1999) were constructed using NETWORK version 4.5.6 (http://fluxusengineering.com) to determine the genetic relationships among the clades that were observed from the NJ tree.

To explore the genetic relationships of the haplotypes generated in this study and those of other chicken populations from different geographic regions and from areas close to the possible centres of chicken domestication, a total of 830 sequences were retrieved from Genbank. The sources and accession numbers of these sequences and their geographic regions were previously described by Liu *et al.* (2006).

Genetic diversity and population structuring

Haplotype and nucleotide diversities were calculated using ARLEQUIN version 3.1 (Excoffier *et al.* 2006). All 113 sequences of the Dutch breeds were considered only in one group according to their original populations, while the 160 sequences of commercial chickens were grouped into different sub-populations to study the extent of genetic structuring in the populations. Analysis of molecular variance (AMOVA) was computed and variance components were estimated among and within the population groups.

Results

Sequence variation and haplotype sharing

A total of 20 haplotypes (HM015602-21) were identified in this study (Table S1). Twelve haplotypes were identified in the fancy breeds, while 14 haplotypes were identified in the commercial chickens. In total, 31 substitutions (two of which were transversions) were detected. Information on the samples and the distribution of haplotypes are shown in Table 1. Six dominant haplotypes (A2, B1, E1, E5, E7 and E10) were present in 222 chickens (of 273). Except for haplotypes E1 and E7, <3 individuals of fancy chickens shared haplotypes with commercial chickens. Six haplotypes found in 16 samples (of 113) were specific to fancy chickens, while eight haplotypes, found in 56 chickens (of 160), were unique to commercial samples (Fig. 1).

Phylogenetic relationships and network analysis

Phylogenetic analysis of the sequences revealed five clades. Except for one clade (D) that belonged to clade F, they exactly correspond to clades A through E as defined by Liu *et al.* (2006), and we retained the same nomenclature throughout this paper. Clade A contained five haplotypes. Clades B, C and D comprise one haplotype each and represent a small number of individuals. Clade E is the major clade widely distributed among all breeds (Table 2). Twelve haplotypes representing 224 (of 273) samples belong to this clade (Fig. 2). The phylogenetic clades in the NJ tree were supported by the MJ network (Fig. 3).

	16 14	16 41	17 16	19 46	20 47	20 79	21 40	21 94	22 19	22 92	23 13	23 36	23 69	24 03	24 36	25 13	25 38	26 58	27 01	27 24	31 17	32 47	33 79	34 91	35 10	36 00	38 48	Fancy	BM	BF	BE	WE	Chi na	Jap ani	Ind ia	Indo nes ia	Sou the ast	Mid dle ast	Eur ope
ST	T	C	T	T	A	T	G	T	A	T	C	T	A	G	T	C	A	T	T	C	A	A	C	C	A	C	A	A	A	C	T	C									
A1	·	·	·	·	·	C	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	1	0	0	5	0	†	†	–	–	–	†	†
A2	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	2	0	0	7	1	†	†	–	–	–	†	†
A3	·	·	·	·	·	C	·	·	·	C	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	2	0	0	0	0	†	–	–	–	–	–	–
A4	·	·	·	·	·	C	·	·	·	·	·	·	G	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	0	4	2	0	0	–	–	–	–	–	–	–
A5	C	·	C	·	·	C	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	0	0	0	1	0	–	–	–	–	–	–	–
B1	·	T	·	·	·	C	A	·	·	C	·	·	·	·	·	·	T	·	·	·	·	·	·	·	T	·	·	·	·	·	·	·	0	9	6	0	0	†	†	†	†	†	†	–
C1	·	T	·	·	·	C	A	·	·	C	·	·	·	·	A	C	·	·	·	C	·	·	G	T	·	·	·	·	·	·	T	C	·	6	0	0	0	0	†	†	–	–	–	–	–
D1	·	T	·	·	·	C	·	·	·	C	T	C	·	·	C	·	·	C	C	T	·	·	T	T	C	·	·	·	G	·	·	·	3	0	0	0	0	–	–	–	–	–	–	–
E1	·	T	·	·	·	C	·	C	·	C	·	·	·	·	C	·	·	·	C	T	·	·	T	·	·	·	·	·	·	·	·	·	80	3	3	9	0	†	†	†	–	–	†	†
E2	·	T	·	·	·	C	·	C	·	C	·	·	·	·	C	·	·	·	C	T	·	·	T	·	·	·	·	G	·	·	·	·	2	0	0	0	0	–	–	–	–	–	–	–
E3	·	T	·	·	·	C	·	C	·	C	·	·	·	·	C	·	·	·	C	T	·	·	T	·	·	T	·	·	·	·	·	·	2	0	0	5	0	†	–	–	–	–	†	–
E4	·	T	·	·	·	C	·	C	·	C	·	·	·	·	C	·	G	·	C	T	·	·	T	·	·	·	·	·	·	·	·	·	3	0	0	0	0	–	–	–	–	–	–	–
E5	·	T	·	C	·	C	·	C	·	C	·	·	·	·	C	·	·	·	C	T	·	·	T	·	·	·	·	·	·	·	·	·	2	19	45	0	0	†	†	–	†	†	–	†
E6	·	T	·	·	G	C	·	C	·	C	·	·	·	·	C	·	·	·	C	T	·	·	T	·	·	·	·	·	·	·	·	·	0	0	1	0	0	–	–	–	–	–	–	–
E7	·	T	·	·	·	C	·	C	·	C	·	·	·	·	C	T	·	·	C	T	·	·	T	·	·	T	·	·	·	·	·	·	9	0	0	0	7	–	†	–	–	–	†	–
E8	·	T	·	·	·	C	·	C	·	C	·	·	·	·	C	·	·	·	C	T	·	·	T	·	·	·	G	·	·	·	·	·	0	0	0	4	1	†	–	–	–	–	–	†
E9	·	T	·	C	·	C	·	C	·	C	·	·	·	·	C	·	·	·	C	T	G	·	T	·	·	·	·	·	·	·	·	·	0	1	0	0	0	–	–	–	–	–	–	–
E10	·	T	·	·	·	C	·	C	G	C	·	·	·	·	C	·	·	·	C	T	·	·	T	·	·	·	·	·	·	·	·	·	0	0	0	20	0	–	–	–	–	–	–	†
E11	·	T	·	C	·	C	·	C	·	C	·	·	·	·	C	·	·	·	C	·	·	·	T	·	·	·	·	·	·	·	·	·	1	0	0	0	0	–	–	–	–	–	–	–
E12	·	T	·	·	·	C	·	C	·	C	·	·	·	·	C	·	·	·	C	T	·	·	T	·	·	·	·	·	·	·	·	A	0	0	0	0	7	–	–	†	–	–	–	–

Figure 1 Sequence variations of 20 mtDNA haplotypes (HM015602-HM015621) of 113 samples of Dutch fancy and 160 samples of commercial chicken populations. The reference sequence used here (ST) is the mtDNA fragment between regions 30–394 of *Gallus gallus domesticus*, X52392 (Desjardins & Morais 1990). Insertions were excluded from analysis. Nucleotide positions identical to the reference sequence are indicated by dots. The numbers under the different populations/breeds denote the number of individuals observed for the haplotypes. Marks in the last column show the presence (+) or absence (–) of our haplotypes in each region based on comparison of our haplotypes to 830 sequences of domestic chickens used by Liu *et al.* (2006). BM, combined broiler male lines 1 & 3; BF, combined broiler female lines 2 & 4; BE, brown-egg layer; WE, white-egg layer.

Table 2 Diversity of Dutch fancy and commercial populations of chickens and distribution of individuals in the clades.

Breed[1]	Haplotype diversity (±SE)	Nucleotide diversity (±SE)	Distribution of individuals in each clade, *n* (%)[2]				
			A	B	C	D	E
Fancy							
GrM	0.524 (0.209)	0.0021 (0.002)					8 (100)
Lak	0.533 (0.172)	0.0101 (0.007)			2 (33)		4 (67)
DrF	0.000	0.000					8 (100)
AsF	0.000	0.000					8 (100)
FrF	0.600 (0.175)	0.0098 (0.007)	2 (40)				3 (60)
Ham	0.533 (0.172)	0.0029 (0.003)					8 (100)
PoB	0.000	0.000					8 (100)
DoB	0.000	0.000					8 (100)
PnB	0.000	0.000					8 (100)
BrF	0.750 (0.097)	0.0198 (0.012)	2 (25)			3 (38)	3 (38)
Bra	0.607 (0.164)	0.0018 (0.002)					8 (100)
DuB	0.524 (0.209)	0.0067 (0.005)	1 (14)				6 (86)
BoB	0.571 (0.095)	0.0109 (0.007)			4 (50)		4 (50)
Bar	0.000	0.000					8 (100)
HoF	0.000	0.000					8 (100)
Kra	0.000	0.000					8 (100)
All Fancy	0.490 (0.057)	0.0063 (0.003)	5 (4)		6 (5)	3 (3)	99 (88)
Commercial							
BM1	0.467 (0.148)	0.0067 (0.004)		2 (13)			13 (87)
BF2	0.371 (0.153)	0.0042 (0.003)		1 (7)			14 (93)
BM3	0.733 (0.052)	0.0149 (0.008)	4 (19)	7 (33)			10 (48)
BF4	0.373 (0.089)	0.0073 (0.004)	2 (5)	5 (12)			35 (83)
BE	0.786 (0.039)	0.0113 (0.006)	13 (25)				38 (75)
WE	0.650 (0.075)	0.0059 (0.003)	1 (6)				15 (94)
B1–2	0.409 (0.108)	0.0053 (0.003)		3 (10)			27 (90)
B3–4	0.536 (0.062)	0.0110 (0.006)	6 (10)	12 (19)			45 (69)
BM1–3	0.657 (0.063)	0.0134 (0.007)	4 (11)	9 (25)			23 (64)
BF2–4	0.368 (0.077)	0.0061 (0.004)	2 (3)	6 (11)			49 (86)
B	0.497 (0.056)	0.0102 (0.005)	6 (7)	15 (16)			72 (77)
L	0.853 (0.024)	0.0113 (0.006)	14 (21)				53 (79)
Com	0.801 (0.026)	0.0112 (0.006)	20 (13)	15 (9)			125 (78)
Fancy-Com	0.807 (0.016)	0.0092 (0.005)	25 (9)	15 (6)	6 (2)	3 (1)	224 (82)

[1]Abbreviations for breeds are given in Table 1.

[2]Number of individuals in the clade, figures in parenthesis indicate percentage of individuals in the clade out of the total number of samples for the breed.

Clades A and E contain most of the haplotypes. Clade A has one central haplotype, A1, with three others distributed around it at a maximum of two mutational distances. The haplotypes in this clade are mainly found in commercial breeds. Haplotypes A1, A2 and A3 in this analysis are the same as haplotypes A1, A2 and A7 described by Liu *et al.* (2006) (Table S1, Fig. S1). Haplotypes A1 and A2 were found in brown-egg layers and in a single individual of Dutch bantam (A1) and two Breda fowl animals (A2). Haplotypes A4 and A5 are uniquely found in this study. The broiler haplotype (B1) and haplotype C1, containing Lakenvelder and Booted bantam animals, are the same as haplotype B1 and C6 described by Liu *et al.* (2006). Haplotype E1, the major E-clade haplotype, was also described by Liu *et al.* (2006). All the other haplotypes in the clade were distributed around it with one mutation difference, except for E9, E10 and E11, which were separated by two differences from E1 (Fig. 3). The haplotypes E3, E5, E7, E8, E10 and E12 correspond to haplotypes E3, E6, E8, E9, E11 and E15 previously described by Liu *et al.* (2006).

Eight of the haplotypes identified in this study were not found by Liu *et al.* (2006). Four haplotypes of these eight belong to three fancy breeds (D1 and E4 found in six Breda fowls, E2 found in two Hamburgh fowls and E11 found in one Dutch bantam). The other four novel haplotypes were found in commercial breeds (A4 in four individuals from broiler male line 3 and two individuals from broiler female line 4; A5 found in one brown-egg layer; E6 and E9 found in one individual each from broiler female line 2 and broiler male line 1).

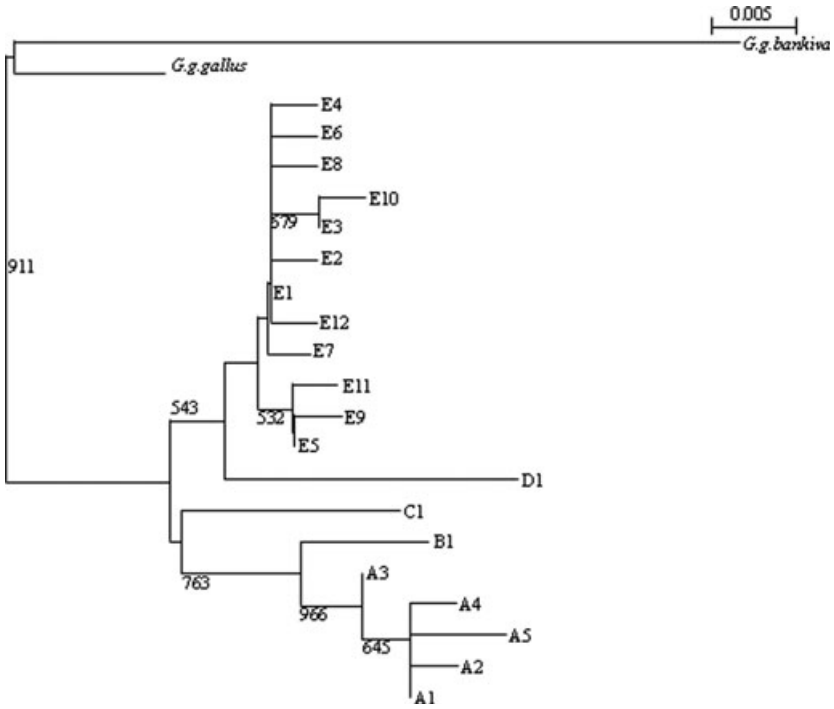


Figure 2 A neighbour-joining tree for the 20 haplotypes of mtDNA control region sequences of 113 Dutch fancy and 160 commercial chickens. The CLUSTALX package was used to draw the tree applying 1000 bootstrap replications to the b test reliability of the branching order. The red jungle fowls (*Gallus gallus bankiva*, AB007718 and *G. g. gallus*, AB007720) were used as outgroups to root the tree using NJPLOT95. Only bootstrap values >50% are presented.

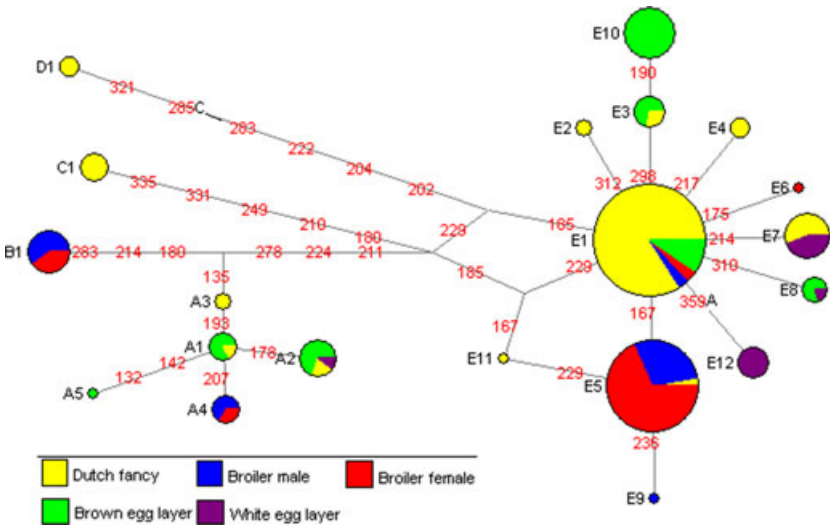


Figure 3 Median-joining network among 20 mtDNA haplotypes of Dutch fancy and commercial breeds of chicken based on 365 bp of control-region sequences, regions 30–394 of the reference sequence, *Gallus gallus domesticus*, X52392 (Desjardins & Morais 1990). Suffixes A and C to the nucleotide positions 359 and 285, respectively, designate transversions. Circle areas are proportional to haplotype frequencies.

Genetic diversity

Eight (of 16) fancy breeds belonged to a single haplotype, E1. Equal proportions of the remaining eight breeds have 2 and 3 haplotypes. The haplotype diversity in these breeds ranged from 0.52 (Groninger Mew and Dutch bantam) to 0.75 (Breda fowl), while the nucleotide diversity values ranged between 0.0018 (Brabanter) and 0.0198 (Breda fowl). Each of the commercial breeds belonged to four haplotypes, except the brown-egg layers, which belonged to 7 (Table 1). The haplotype and nucleotide diversities of commercial breeds ranged, respectively, from 0.37 to 0.78 and 0.0042 to 0.0149. All fancy chickens considered together showed haplotype diversity of 0.49 and nucleotide diversity of 0.0063; the respective figures for the commercial populations were 0.80 and 0.0112 (Table 2).

Population structure

Estimates of a global AMOVA for Dutch fancy chickens, treating all 16 breeds as one group, indicated that 72.5% ($P < 0.0001$) of the genetic variance was because of variation within populations. At the same time, the diversity among breeds (27.5%) was also found to be highly significant ($P < 0.0001$). Likewise, significant genetic variation was observed both within (80%, $P < 0.0001$) and among (20%, $P < 0.0001$) the breeds in commercial populations,

when all breeds were treated as a single group. The variation among the fancy and commercial populations (5.8%) was non-significant ($P > 0.05$). When commercial populations were considered separately and grouped as broilers vs. layers and broiler males vs. broiler females, only 11.3% ($P > 0.05$) of the variation among broilers and layers and 2.4% ($P > 0.05$) of the variation among broiler males and broiler females could be attributed to the genetic differences among the groups. Further grouping of the broilers by the source of the samples, as broilers from company A vs. those from B, indicated that genetic differences because of company groups were confounded by large errors of estimations (Table 3). In general, the diversity among breeds was larger for fancy populations compared to commercial populations. However, a higher level of diversity exists within commercial breeds than in fancy breeds.

Discussion

This study presents a systematic survey of mitochondrial haplotype diversity of nearly all recognized traditional breeds of chicken of the Netherlands. The breeds studied vary in background. Some are thought to originate from country fowl, while others have a known or putative history of contributions of chicken from other parts of Europe or even from other continents. Some fancy breeds such as the Breda fowl, Booted bantam, Dutch owl-bearded, Polish bearded, Polish non-bearded and Hamburgh are thought to be relatively old, featuring in 16th and 17th century paintings, while other Dutch breeds were created in the 20th century to meet new demands for commercial egg and meat production (Appendix S1). Despite the diverse backgrounds of Dutch breeds, the common mitochondrial theme is the E-clade, and specifically the E1 haplotype; all breeds have E-clade haplotypes, and only the Breda fowl and the Hamburgh do not have the E1 haplotype. This clade is the same clade (clade E) described by Liu *et al.* (2006), which mainly contained chickens from Europe, the Middle East and India. Considering the distribution patterns of the clade and the high proportion of unique haplotypes in India, Liu *et al.* (2006) suggested that the origin of this clade might be the Indian subcontinent. The consistent occurrence of this clade in the entire traditional chicken diversity of a Western European country suggests that European chickens are originally mostly if not exclusively derived from the Indian subcontinent.

Apart from the consistent occurrence of the E-clade in Dutch fancy breeds, there is a sporadic occurrence of a few other clades, with only a few of the occurrences being shared between breeds. Three breeds had clade A haplotypes, with none of the haplotypes overlapping between breeds. Haplotypes in clade A were found before among large numbers of Chinese (109) and Japanese (16), and a small number of European (2) and Iranian (2) chickens (Liu *et al.* 2006). Two Dutch breeds (Lakenvelder and Booted bantam) shared the same clade C (C1) haplotype l; this is a group of haplotypes found so far only in Chinese and Japanese chickens. Clade D, found only in Breda Fowl, was clustered in clade F, which was described by Liu *et al.* and in that study contained only samples from Yunnan, South China. The erratic occurrence of East Asian haplotypes in Western European breeds suggests that there is no ancient occurrence of these haplotypes that would have been likely to create a more uniform distribution. Rather, the irregular distribution of these haplotypes appears much more to be the product of the happenstance availability of exotic chickens in combination with the whims and fancies of chicken breeders in past centuries.

As with the Dutch fancy breeds, the dominant haplotypes of the commercial chickens are in the E-clade. All commercial breeds originated in Europe or North America, and particularly, the broilers were thought to have a very high proportion of Asian contributions to their origin. However, from the mitochondrial signatures, even the

Table 3 AMOVA analysis of Dutch fancy and commercial populations of chicken based on mtDNA control region variation.

	Source of variation (%)		
Group (G)	Among groups	Among populations within groups	Within populations
Fancy[1] – no grouping	27.47 (0.000)		72.53 (0.000)
Commercial[2] – no grouping	19.95 (0.000)		80.05 (0.000)
Fancy vs. commercial	5.81 (0.065)	18.07 (0.000)	76.11 (0.000)
Broiler vs. layer (G1 = BM1–3, BF2–4; G2 = BE, WE)	11.30 (0.329)	11.48 (0.000)	77.19 (0.000)
Broiler males vs. broiler females (G1 = BM1, BM3; G2 = BF2, BF4)	2.35 (0.319)	11.15 (0.022)	86.50 (0.003)
Broilers from company A vs. broilers from company B (G1 = BM1, BF2; G2 = BM3, BF4)	−2.94 (0.647)	14.85 (0.006)	88.09 (0.004)

[1]The sub-populations were formed of the 16 fancy breeds (see Table 1).
[2]The sub-populations were formed of BM1–3, BF2–4, BM1, BF2, BM3, BF4, BE, WE (see Table 1 for abbreviations).
Significance tests were based on 1023 permutations; *P* values are indicated in parentheses.

broilers appear to mostly descend from European chickens. Nevertheless, the pattern of E-clade diversity in commercial birds is much more complex, and the broilers have mostly E5; in the brown-egg layers, E10 is dominant and in the white-egg layers, E7 and E12 make up virtually all the variation. The E-clade is nowadays occurring throughout the globe, no doubt largely because of recent translocations by European traders. However, there may be a more ancient dispersal of the E-clade, for instance into Asia, that needs further investigation, and that makes a precise origin of the commercial chicken breeds currently difficult to ascertain.

As in the Dutch fancy breeds, East Asian haplotypes can be found in the commercial breeds. However, unlike in the Dutch fancy breeds, the pattern of occurrence of these haplotypes is much more consistent. The A1 and A2 haplotypes are found mostly in the brown-egg layers (and a single occurrence in a white-egg layer) and are shared with a few Dutch breeds. The occurrence of these A-clade haplotypes, which are found at high frequency in Chinese chickens, is in line with the historical records that suggest a contribution of Cochins and Langshans to brown layers. A unique A-clade haplotype, A4, occurs only in broilers, and interestingly only in broilers of one company, both in dam and sire lines, suggesting cross-breeding of what are considered to be distinct breeds during the history of these commercial lines; this is supported by SNP markers (Megens *et al.* 2009). The A4 haplotype appears therefore to be a rare occurrence in the broilers, and in these commercial chickens the B-clade, specifically the B1 haplotype that is also occurring at high frequency in China and Japan, it appears to be the common theme. This indicates introgression of East Asian chickens of a very different source compared to the brown-egg layers and Rhode Island Red, despite the historical records indicating a common origin from similar dual purpose breeds.

The occurrence of mitochondrial haplotypes that are mostly East Asian in distribution in Western European traditional breeds and commercial broilers and layers confirms historical records that indicate such influences. The occurrence of many different clades indicates that many different Asian breeds may have been used. Although East Asian haplotypes constitute a minority in most populations surveyed here, the per cent contribution of East Asian chicken to current Dutch and commercial chicken populations is difficult to establish from mtDNA alone.

The influence of the introgression on genetic diversity becomes apparent from the nucleotide diversity and haplotype diversity analyses. All populations that had haplotypes of multiple origin displayed high inferred diversity, as opposed to most populations that had only a single origin mitochondrial signature (mostly the E1 haplotype). Interestingly, most fancy breeds have been shown to have a much lower within-population diversity compared to broilers and layers based on nuclear markers (Eding *et al.* 2002), but this is not the case for mitochondrial estimates in fancy breeds that have multiple origin haplotypes. The results of AMOVA showed that there was genetic differentiation among Dutch fancy breeds and also among commercial breeds. The ragged mismatch distribution curve of fancy and commercial populations also supports the presence of population subdivision and the heterogeneity between populations (Table S2, Fig. S2).

Understanding the complex origin of chicken populations is of vital importance for comprehending, preserving and exploiting chicken genetic diversity. A complex origin, in this study shown to occur in broilers, brown-egg layers and several Dutch traditional fancy breeds, may influence nuclear haplotype diversity, linkage disequilibrium, and hence this affects all mapping and association analyses including genomic selection.

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Supporting information

Additional supporting information may be found in the online version of this article.

Figure S1 Median-Joining network of haplotypes observed in this study and haplotypes of this study observed in domestic chicken haplotypes of Liu *et al.* (2006).

Figure S2 Mismatch distribution for mtDNA types of Dutch fancy (a) and commercial (b) populations and broiler (c) and layer (d) breeds.

Table S1 Names and accession numbers of haplotypes defined by Liu *et al.* (2006) that were used to compare with the haplotypes observed in this study.

Table S2 Summary statistics of Dutch fancy and commercial populations of chicken, broilers and layers showing raggedness and Fu's *Fs* values.

Appendix S1 History of Dutch fancy breeds used in this study.

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